## Graph a Plane Curve by Plotting Points

**Plane Curve**

Let $f$ and $g$ be functions of $t$ over an interval $I$. Then the set of points defined by $(x, y) = (f(t), g(t))$

is called a **plane curve**.

$$x = f(t) \qquad y = g(t)$$

**Parametric Equations**

The equations and are called **parametric equations** of the curve, and $t$ is called the **parameter**.

**Graphing a Plane Curve**

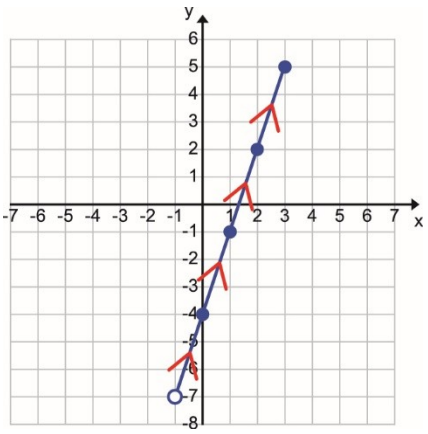
To create a table of points, choose arbitrary values of the parameter $t$ and compute the corresponding values of $x$ and $y$. Plot the points $(x, y)$ in the order of increasing $t$ and connect the points with a curve. Then use arrows to indicate the **orientation** of the curve. This is the direction of the path for increasing values of $t$.

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1. Sketch the curve by plotting points over the interval $-3 < t \le 9$. Place arrows indicating increasing values of $t$.

$x = \frac{t}{3}$ and $y = t - 4$

| $t$ | $x$ | $y$ | |
|---|---|---|---|
| −3 | −1 | −7 | ○ |
| 0 | 0 | −4 | ● |
| 3 | 1 | −1 | ● |
| 6 | 2 | 2 | ● |
| 9 | 3 | 5 | ● |




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## Eliminate the Parameter

Given parametric equations: → Solve for $t$ or an expression involving $t$ in one equation. → Substitute the result in the other equation. → Identify domain restrictions on the resulting rectangular equation.

2. Eliminate the parameter $t$, sketch the curve, and indicate the orientation.

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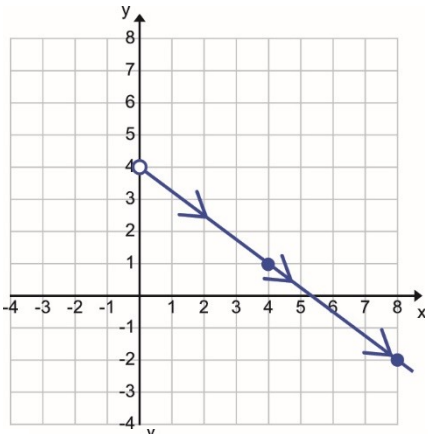
$x = 4e^t,\ y = -3e^t + 4$

$e^t = \frac{x}{4}, \quad y = -3\left(\frac{x}{4}\right) + 4$

$y = -\frac{3}{4}x + 4$

$\text{slope} = -\frac{3}{4}, \quad y - \text{int } (0, 4)$

$x > 0$

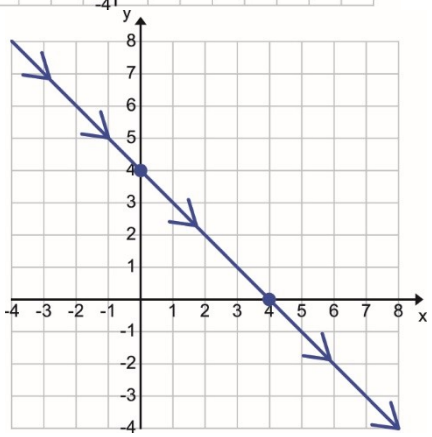



3. Eliminate the parameter $t$, sketch the curve, and indicate the orientation.

$x = 8t,\ y = -6t + 4 \qquad t = \frac{x}{8}, \quad y = -6\left(\frac{x}{8}\right) + 4$

$y = -\frac{3}{4}x + 4$

no restrictions on $x$




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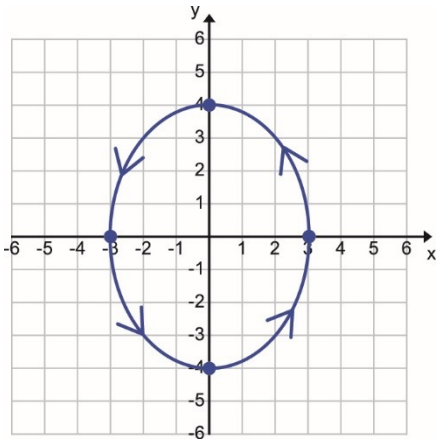
4. Eliminate the parameter $t$, sketch the curve, and indicate the orientation.

$x = 3\cos t,\ y = 4\sin t,\ 0 \le t < 2\pi$

$-3 \le x \le 3, \quad -4 \le y \le 4$

$\cos t = \frac{x}{3}, \quad \sin t = \frac{y}{4} \quad \cos^2 t + \sin^2 t = 1$

$\frac{x^2}{9} + \frac{y^2}{16} = 1$




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## Write Parametric Equations to Represent a Curve

Write parametric equations for the curve for the given definitions of $x$.

5. $y = 5x - 2$

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a.

$x =$

$y =$

$y =$

$x =$

$x = 2t + 5$

$y = 5(2t + 5) + 5$

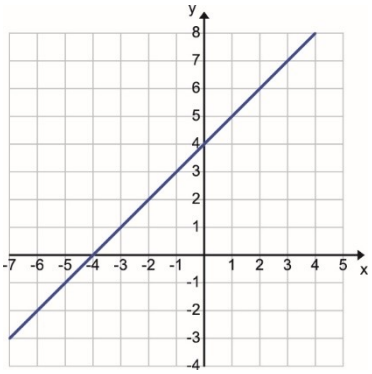
$= 10t + 25 + 5$

$y = 10t + 30$

b. $x = 2t + 5,\ y = 10t + 30$

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6. Write parametric equations for $t$ to define the given graph. There may be multiple solutions.




$$\left.\begin{array}{l} y\text{-intercept} = (0,4) \\ \text{slope} = 1 \end{array}\right\} y = x + 4 \rightarrow \begin{array}{l} x = t \\ y = t + 4 \end{array}$$

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Apply Parametric Equations

| Modeling Motion Using Parametric Equations | |
|---|---|
| Let $t \geq 0$ represent the time under which an object is in motion along a plane curve, and let $(x_0, y_0)$ represent the initial position of the object at $t = 0$. | |
| **Uniform linear motion**<br>$x = x_0 + at$<br>$y = y_0 + bt$ | The values of $a$ and $b$ are the horizontal and vertical components of velocity, respectively. |
| **Projectile motion***<br>$x = (v_0 \cos\theta)t + x_0$<br>$y = -\frac{1}{2}gt^2 + (v_0 \sin\theta)t + y_0$<br>*Assuming negligible air resistance. | $\theta$ is the angle of launch relative to the horizontal, and $v_0$ is the initial speed (magnitude of the initial velocity).<br>The value of $g$ is the acceleration due to gravity,<br>$g = 32$ ft/sec$^2$ or $g = 9.8$ m/sec$^2$<br>where on the surface of the Earth. |

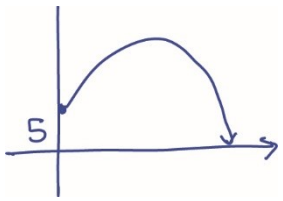
7. An object undergoes uniform linear motion on a straight path from point $A$ (3, 8) to point $B$ (6, 10) in 3 seconds. Write parametric equations over an interval $I$ that describes the motion along the path.

$x_0 = 3, y_0 = 8, t = 3,$ $a = \frac{6-3}{3} = 1,$ $b = \frac{10-8}{3} = \frac{2}{3}$

$x = x_0 + at$ $y = y_0 + bt$

$x = 3 + t$ $y = 8 + \frac{2}{3}t$

8. A ball is thrown from a height of 5 feet at a speed of 80 ft/sec and an angle of 60° from the horizontal. Write parametric equations to describe the path of the ball as a function of time, $t$. Determine when the ball hits the ground.




$x_0 = 0, y_0 = 5, \theta = 60°, v_0 = 80$

$x = (80\cos 60°)t + 0$

$x = 40t$

$y = -\frac{1}{2}(32)t^2 + (80\sin 60°)t + 5$

$y = -16t^2 + 40\sqrt{3}t + 5$

when ball hits the ground, $y = 0$

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$$t = \frac{-40\sqrt{3} \pm \sqrt{(40\sqrt{3})^2 - 4(-16)(5)}}{2(-16)}$$

$-16t^2 + 40\sqrt{3}t + 5 = 0$ $t \approx -0.07$, 4.40 seconds

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